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COURSE NAME : Problem solving and testing using java

COURSE CODE : 10211CS227

TASK2:

Task 2 (Difficult): Real-Time Stream Analytics Engine

PROBLEM DESCRIPTION:

A cloud analytics platform receives millions of sensor readings. Each reading contains a sensor ID and temperature value. Using stream processing concepts, perform the following operations:

1. Filter temperatures greater than 50.
2. Group readings by sensor ID.
3. Compute average temperature per sensor.
4. Sort sensors based on average temperature in descending order.

Input Format

- First line contains integer N.
- Next N lines contain SensorID and Temperature.

Output Format

Display SensorID and average temperature sorted in descending order.

Constraints

- $1 \leq N \leq 10^5$
- $0 \leq \text{Temperature} \leq 100$

Sample Input

6

S1 60

S2 40

S1 80

S3 70

S2 90

S3 30

Sample Output

S1 70.0

S2 90.0

S3 70.0

PROGRAM:

```
import java.util.*;

public class Main {

    public static void main(String[] args) {

        Scanner in = new Scanner(System.in);

        int n = in.nextInt();

        // SensorID -> [sum of temperatures, count]
        Map<String, double[]> map = new HashMap<>();

        for (int i = 0; i < n; i++) {

            String sensorId = in.next();

            double temperature = in.nextDouble();

            // Filter: temperature > 50
            if (temperature > 50) {

                if (!map.containsKey(sensorId)) {

                    map.put(sensorId, new double[]{0, 0});

                }

                map.get(sensorId)[0] += temperature; // sum
                map.get(sensorId)[1]++;             // count
            }
        }

        List<Map.Entry<String, double[]>> list =

            new ArrayList<>(map.entrySet());
```

```

list.sort((a, b) -> {
    double avgA = a.getValue()[0] / a.getValue()[1];
    double avgB = b.getValue()[0] / b.getValue()[1];
    return Double.compare(avgB, avgA);
});
for (Map.Entry<String, double[]> entry : list) {
    String sensorId = entry.getKey();
    double sum = entry.getValue()[0];
    double count = entry.getValue()[1];
    double average = sum / count;
    System.out.println(sensorId + " " + average);
}
in.close();
}
}

```

OUTPUT:

```

6
S1 60
S2 40
S1 80
S3 70
S2 90
S3 30
S2 90.0
S3 70.0
S1 70.0

```